

Question number	Answer	Additional guidance	Mark
8(a)(i)	A description that includes three of the following points: • need to withstand {harsh / extreme} environments (1) • need to be able to withstand {dessication / lack of shade} (1) • have a low requirement for {minerals / mineral ions / named mineral ion / soil} (1) • have {fast life-cycles / grow fast / reproduce asexually / produce lots of seeds / wide dispersal mechanisms} (1)	ACCEPT can live in dry conditions ACCEPT can grow in {poor quality soil / bare rock} IGNORE nutrients	(3)

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8(a)(ii)	An explanation that includes two of the following points: • changes to the habitat have to take place before a different organism can survive there (1) • credit named improvement (1) • credit organism that would appear in next stage following this improvement (1)	ACCEPT (one species) {improves conditions / provides suitable conditions} (for the next species) e.g. make soil, improve soil, provide food (for animals) e.g. (deeper soil) bushes / trees, (food / shelter) animals	(2)

Question number	Answer	Additional guidance	Mark
8(b)(i)	An answer that includes the following points: • eruption results in a drop in temperature (within 1.25 years) (1) • by 0.2°C (1)	IGNORE references to subsequent increase ACCEPT value in range of 0.02 to 0.2, but must be correct if time period stated OR • (overall) increase in temperature (1) • by {0.1 / 0.15}°C (1)	(2)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	An explanation that includes the following points: - less (UV / shorter wave length) light will (indirectly) result in cooler temperatures (1) - because there will be less {IR (radiation) / long wave length} (reflected from the Earth's surface to warm up the atmosphere) (1) - with time, the particles will {disperse / decrease} (1) - and more (UV / shorter wave length) light will be able to {reach the Earth / pass through} increasing the temperature (1)	ACCEPT {particles / ash / sulfur dioxide} throughout IGNORE heat ACCEPT less heat trapped	(4)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	700 / 7×10^2 (times / eruptions) (1)		(1)

Question number	Answer	Additional guidance	Mark
8(b)(iv)	An answer that includes four of the following points: - graph shows that there is no significant change in levels of carbon dioxide following the eruptions (1) - but there is evidence that volcanoes release some carbon dioxide (1) - so they might contribute to the greenhouse effect and cause {climate change / global warming / increase in temperatures} (1) - but this might be offset by the {sulfur dioxide / ash} produced (1)	ACCEPT volcanoes {do release / increase} carbon dioxide ACCEPT volcanoes do contribute to the greenhouse effect and cause {climate change / global warming / increase in temperatures} ACCEPT carbon dioxide is a greenhouse gas and causes climate change	(3)

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